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Topic 10.1: Bisection Method (Examples)

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Example 1

Consider finding the root of $f(x) = x^2 - 3$. Let $\varepsilon_{\text{step}} = 0.01$, $\varepsilon_{\text{abs}} = 0.01$ and start with the interval $[1, 2]$.

Table 1. Bisection method applied to $f(x) = x^2 - 3$.

a	b	$f(a)$	$f(b)$	$c = (a + b)/2$	$f(c)$	Update	new $b - a$
1.0	2.0	-2.0	1.0	1.5	-0.75	$a = c$	0.5
1.5	2.0	-0.75	1.0	1.75	0.062	$b = c$	0.25
1.5	1.75	-0.75	0.0625	1.625	-0.359	$a = c$	0.125
1.625	1.75	-0.3594	0.0625	1.6875	-0.1523	$a = c$	0.0625
1.6875	1.75	-0.1523	0.0625	1.7188	-0.0457	$a = c$	0.0313
1.7188	1.75	-0.0457	0.0625	1.7344	0.0081	$b = c$	0.0156
1.71988	1.7344	-0.0457	0.0081	1.7266	-0.0189	$a = c$	0.0078

Thus, with the seventh iteration, we note that the final interval, $[1.7266, 1.7344]$, has a width less than 0.01 and $|f(1.7344)| < 0.01$, and therefore we chose $b = 1.7344$ to be our approximation of the root.

Example 2

Consider finding the root of $f(x) = e^{-x}(3.2 \sin(x) - 0.5 \cos(x))$ on the interval $[3, 4]$, this time with $\varepsilon_{\text{step}} = 0.001$, $\varepsilon_{\text{abs}} = 0.001$.

Table 1. Bisection method applied to $f(x) = e^{-x}(3.2 \sin(x) - 0.5 \cos(x))$.

a	b	$f(a)$	$f(b)$	$c = (a + b)/2$	$f(c)$	Update	new $b - a$
3.0	4.0	0.047127	-0.038372	3.5	-0.019757	$b = c$	0.5
3.0	3.5	0.047127	-0.019757	3.25	0.0058479	$a = c$	0.25
3.25	3.5	0.0058479	-0.019757	3.375	-0.0086808	$b = c$	0.125
3.25	3.375	0.0058479	-0.0086808	3.3125	-0.0018773	$b = c$	0.0625
3.25	3.3125	0.0058479	-0.0018773	3.2812	0.0018739	$a = c$	0.0313
3.2812	3.3125	0.0018739	-0.0018773	3.2968	-0.000024791	$b = c$	0.0156
3.2812	3.2968	0.0018739	-0.000024791	3.289	0.00091736	$a = c$	0.0078
3.289	3.2968	0.00091736	-0.000024791	3.2929	0.00044352	$a = c$	0.0039
3.2929	3.2968	0.00044352	-0.000024791	3.2948	0.00021466	$a = c$	0.002
3.2948	3.2968	0.00021466	-0.000024791	3.2958	0.000094077	$a = c$	0.001
3.2958	3.2968	0.000094077	-0.000024791	3.2963	0.000034799	$a = c$	0.0005

Thus, after the 11th iteration, we note that the final interval, $[3.2958, 3.2968]$ has a width less than 0.001 and $|f(3.2968)| < 0.001$ and therefore we chose $b = 3.2968$ to be our approximation of the root.

Example 3

Apply the bisection method to $f(x) = \sin(x)$ starting with $[1, 99]$, $\epsilon_{\text{step}} = \epsilon_{\text{abs}} = 0.00001$, and comment.

After 24 iterations, we have the interval $[40.84070158, 40.84070742]$ and $\sin(40.84070158) \approx 0.0000028967$. Note however that $\sin(x)$ has 31 roots on the interval $[1, 99]$, however the bisection method neither suggests that more roots exist nor gives any suggestion as to where they may be.

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